

Exercises from chapter 4

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Exercise 4

Let's define first $H_{D0}^1(\Omega) = \{v \in H^1(\Omega) : v|_{\Gamma_D} = 0\}$. Take $v \in H_{D0}^1$. Multiplying it in both sides and integrating, as seen previously, we obtain:

$$\int_{\Omega} \mu \nabla u \cdot \nabla v dx - \int_{\Gamma_D \cup \Gamma_R} \mu \nabla u \cdot \tilde{n} v ds = \int_{\Omega} f v dx \quad (1)$$

Analysing the boundary term, noticing that $v = 0$ at Γ_D :

$$\int_{\Gamma_D \cup \Gamma_R} \mu \nabla u \cdot \tilde{n} v ds = \int_{\Gamma_R} \mu \nabla u \cdot \tilde{n} v ds = \int_{\Gamma_R} (h - \gamma u) v ds$$

Using a lift in u , i.e. $u = u_0 + u_g$ with $u_0 \in H_{D0}^1$ and $u_g|_{\Gamma_D} = g$, we can rewrite 1 as

$$\int_{\Omega} \mu \nabla (u_0 + u_g) \cdot \nabla v dx - \int_{\Gamma_R} (h - \gamma(u_0 + u_g)) v ds = \int_{\Omega} f v dx$$

So we have the variational formulation:

Find $u_0 \in H_{D0}^1$ such that $\forall v \in H_{D0}^1$:

$$\int_{\Omega} \mu \nabla u_0 \cdot \nabla v dx + \int_{\Gamma_R} \gamma u_0 v ds = \int_{\Omega} f v dx - \int_{\Omega} \mu \nabla u_g \cdot \nabla v dx + \int_{\Gamma_R} (h - \gamma u_g) v ds$$